Deep Reinforcement Learning for Traffic Light Control in Jönköping

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***Index Terms*—component, formatting, style, styling, insert.**

I. Introduction

A. Motivation

B. Structure of this paper

- What are SUMO limitations? - Do we train on multiple scenarios, like "rush hours" and low traffic conditions, or do we only have normal traffic? - Do we use DQN or are there also other choices like PPO (is PPO able to combine with neural nets?)? - do we make use of curriculum learning? - how do our neural networks actually look like? what is the input, what shape do they have, and what is the output?

a good starting tutorial on how to generate traffic scenarios can be found here [1].

- at the end, compare normal (static) traffic light with our solution. if possible, the static traffic light control should be decently good. maybe its possible to use a real example?

II. Background

A. SUMO

B. Proximal Policy Optimization

as a policy-gradient method, proximal policy optimization (PPO) learns only from the recent experiences that the agent collected.

C. Deep Q Networks

A Deep Q-Network (DQN) is a reinforcement learning algorithm that combines traditional Q-learning with deep neural networks.

III. Related Works

IV. Environment

A. Network

map of sweden downloaded here: https://download.geofabrik.de/europe/sweden.html geojson polygon object created here: https://geojson.io/ chosen geojson found here[1].

Fig. 1. The filter (in grey) that defines the boundaries of the road network in Jönköping.

then, we used osmium and osmfilter to apply the polygon shape to the map of sweden. afterwards, we filter only for highways, except for those of the following types:

- footway
- path
- pedestrian
- steps
- cycleway
- bridleway

finally, we use the SUMO *netconvert* tool to convert the map into a SUMO network.

as for limitations, we extract lanes including speed limit, intersections, and traffic lights. however, since we only model car traffic and not pedestrians, all pedestrian traffic lights will behave completely deterministic. this does not represent the real world where pedestrians can press a button on the traffic light to trigger a faster phase switching.

[1]https://github.com/benbekir/traffic-rl/blob/main/simulations/networks/jkpg/artifacts/jkpg.geojson

Fig. 2. The network as shown in the SUMO nededit UI, highlighting the traffic light placements.

1) Demand Modeling: since there is no publicly available data regarding traffic assignment zones and demand in jönköping, we use the SUMO tool randomTrips [2] to randomly create trips. we create 3600 individual trips in total where a new trip is spawned into the simulation on every timestep.

2) Rush Hours and Noise: im not sure about this yet

B. Observations

- lane density: density is calculated on the entire lane, by considering how many cars are present in the lane and how many cars could potentially fit in the lane. https://github.com/LucasAlegre/sumo-rl/blob/main/sumo_rl/environment/traffic_signal.py#L297

- queue information: queue is calculated by counting the number of cars that are stopped at the traffic light, waiting for the green signal. https://github.com/LucasAlegre/sumo-rl/blob/main/sumo_rl/environment/traffic_signal.py#L309

C. Actions

D. Reward

- predefined reward functions: https://github.com/LucasAlegre/sumo-rl/blob/main/sumo_rl/environment/traffic_signal.py#L347

V. Experiments

TABLE I
OVERVIEW OF THE TRAFFIC SIGNAL CONTROL SYSTEM.

Parameter	**Description**
Method	Deep Q-Network (DQN) and Proximal Policy Optimization (PPO)
Technique	Curriculum Learning with three traffic density stages
	(0.75, 0.875, and 1.0 scale factors)
Scenario	Real-world road network of Jönköping, Sweden
Platform	Simulation of Urban Mobility (SUMO)

Ben choose one of the graph.

[2] https://sumo.dlr.de/docs/Tools/Trip.html#randomtripspy

TABLE II
OVERVIEW OF THE TRAFFIC SIGNAL CONTROL SYSTEM.

Parameter	**Description**
Method	Deep Q-Learning (DQN) and Proximal Policy Optimization (PPO)
Technique	Curriculum Learning with parameters and three traffic density stages (0.75, 0.875, and 1.0 scale factors)
Scenario	Jönköping city center road network (multi-intersection)
Platform	SUMO (Simulation of Urban Mobility)
Agent Type	Multi-Agent with shared policy (parameter sharing)
Reward Function	Queue-based penalty

VI. Results

VII. Conclusion

References

[1] D. A. Guastella, E. Montero-Porras, A. Morales-Hernández, and G. Bontempi, "Traffic modeling with sumo: a tutorial," 2025. [Online]. Available: https://arxiv.org/abs/2304.05982